Supplemental Material:
What a radical-pair quantum reservoir would require

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Every number below is computed under the single protocol used in the main text — correlated-pair birth, converged time grid, out-of-sample capacity estimation — so that no quantity is carried over from a superseded calculation. The sections cover (S1) validation of the Liouville-space solver, (S2) the capacity estimator and its controls, (S3) the injection map and why a single-electron reset fails, (S4) the chemically accessible readout routes and their grid convergence, (S5) the coherence fraction at matched hopping rate, (S6) classical baselines, (S7) the turnover clock and the register-reuse criterion, and (S8) the spin parameters. All are reproducible from the open `qbscreen` package.

S1. LIOUVILLE-SPACE SOLVER AND THE COHERENT LIMIT

The propagator is $e^{\mathcal{L}\tau}$ of the Haberkorn master equation assembled in Liouville space with the column-stacking convention $\mathrm{vec}(A\rho B) = (B^\top \otimes A)\,\mathrm{vec}(\rho)$; the time-integrated singlet yield follows from a single linear solve,

$$\Phi_S = k_S\,\mathrm{vec}(P_S)^\dagger\,(-\mathcal{L})^{-1}\,\mathrm{vec}(\rho_0). \tag{S1}$$

Adding the decoherence term must not corrupt the coherent dynamics. For $1/T_2^e \to 0$ with $k_S = k_T = k$, Eq. (S1) must reproduce the closed-form eigenbasis result $\Phi_S^{\mathrm{coh}} = k\big[\sum_m P_{mm}\rho_{mm}/k + \mathrm{Re}\sum_{m\neq n} P_{mn}\rho_{nm}/(k+i\omega_{mn})\big]$. For a two-electron-plus-nucleus model at $B = 0.9$ mT, $A = 200$ MHz and $k = 1\ \mu\mathrm{s}^{-1}$ the two agree to better than 10^{-6}; this is asserted as a unit test.

S2. CAPACITY ESTIMATOR AND ITS CONTROLS

For a target y_k of the input history the capacity is the out-of-sample squared multiple correlation: a ridge readout ($\lambda = 10^{-6}$) is fitted on the first half of the run and scored on the held-out second half. The totals are $\mathrm{MC} = \sum_{d\geq 0} C[s_{k-d}]$ and $\mathrm{IPC} = \mathrm{MC} + \sum_{\mathrm{nonlin}} C[y]$ over the degree-2 Legendre family (diagonal targets to $d_{\max} = 14$, cross targets for $d_1 < d_2 \leq 6$).

Null floor. Scoring the same feature matrices against targets built from an *independent* input sequence, for which the true capacity is zero, returns $+0.010$ for the estimator used here. The same sum without the $\max(0,\cdot)$ clip returns -1.49, i.e. it is far more biased, in the opposite direction. We therefore retain the clip and quote the floor rather than removing it.

Convergence. The capacity is converged in the length L of the driving sequence (six seeds, engineered point): $\mathrm{IPC} = 5.28 \pm 0.26$, 5.60 ± 0.20, 5.87 ± 0.19 and 5.84 ± 0.10 for $L = 600$, 1200, 2400 and 4800, i.e. flat beyond $L \approx 2400$. Under the superseded single-electron-reset protocol the same estimator did *not* converge, rising from 5.9 to 8.1 over the same range; convergence is a property of the corrected drive, not of the estimator.

The Dambre bound is not an independence test. $\mathrm{IPC} \leq n_{\mathrm{obs}}$ holds asymptotically for an orthonormal target family but is violated in finite samples: for $X = s$ ($n_{\mathrm{obs}} = 1$, $T = 1200$) the estimator returns 1.011 on average and exceeds the bound in 30 of 30 seeds. Satisfying the inequality therefore does not demonstrate that the readout observables are independent, and we make no such argument anywhere in this work.

S3. THE INJECTION MAP

A protocol that resets only the input electron to a product state while retaining the reduced state of the rest is not a chemical process. For a separated pair ($J = 0$) it destroys the electron–electron correlation on every cycle, and since $\langle P_S\rangle = \frac{1}{4} - \langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle$ with $\langle \boldsymbol{S}_2\rangle = 0$, the singlet probability is pinned at exactly 1/4 independently of the input. We replace it by correlated-pair birth: the nuclei are retained and the electron pair is recreated as $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$, or as a coherent S–T_0 superposition. Table S1 gives the consequence at the cryptochrome point ($B = 50\ \mu$T, $J = 0$, $\tau = 1\ \mu$s, six seeds, kinetic-only readout).

TABLE S1. The injection map and the capacity it permits.

injection	max. flux spread	IPC
single-electron reset (superseded)	2.7×10^{-14}	0.000 ± 0.000
correlated-pair birth, S/T	4.9×10^{-1}	2.941 ± 0.108
correlated-pair birth, S–T_0	4.3×10^{-1}	3.116 ± 0.209

Under the superseded map the spread of the observable is at the level of machine precision: the readout is a constant, and the vanishing capacity is an identity of the map rather than a statement about spin chemistry.

S4. CHEMICALLY ACCESSIBLE READOUT ROUTES

Recombination is switched on ($k_S = 1$, $k_T = 0.2\ \mu\text{s}^{-1}$) so that the observable is the yield actually formed, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\,\mathrm{d}t$, sampled at five laboratory times. CIDNP is the product-weighted nuclear polarisation, $\langle I_z \rangle_{\text{prod}} = \int k_S \mathrm{Tr}[I_z P_S \rho]\,\mathrm{d}t \,/ \int k_S \mathrm{Tr}[P_S \rho]\,\mathrm{d}t$: the polarisation carried away by the recombined molecule, not that of the surviving pair.

TABLE S2. Capacity recovered by chemically accessible observables (six seeds).

readout route	channels	IPC
end-point singlet yield	1	1.017
both exit channels, end point	2	1.374
three micro-environments (different τ)	3	2.557
time-resolved kinetics, one product pool	5	2.008
+ accumulated downstream pool	10	2.706
+ CIDNP on the product	8	4.650

Time-grid convergence. A 40 MHz hyperfine coupling has a 25 ns period, so a coarse grid under-samples the dynamics and inflates the time-resolved capacities. Table S3 shows convergence from $n_t = 48$ onward; the main text uses $n_t = 96$ (d$t \approx 10$ ns).

TABLE S3. Convergence in the number of time steps per cycle (six seeds, $L = 700$, the same protocol as Table S2).

n_t	dt (ns)	kinetics IPC	+CIDNP IPC
24	41.7	2.756	4.879
48	20.8	2.050	4.614
96	10.4	2.008	4.650
192	5.2	1.988	4.656

Why CIDNP is not an optional extra channel. The reduced state of either electron of a newly born pair is maximally mixed for every input, because both $|S\rangle\langle S|$ and $P_T/3$ carry zero one-electron polarisation; the input resides entirely in the two-electron correlation. With $J = 0$ the radicals do not interact, so nothing transfers that correlation to a nucleus. Consistently, with recombination switched off the cryptochrome-point capacity is 1.01, the memoryless floor. Spin-selective recombination is what converts the singlet–triplet correlation into a population difference and so writes the input into the nuclear register: in a separated pair it is the write mechanism, not merely the read mechanism.

S5. COHERENCE FRACTION AT MATCHED HOPPING RATE

An all-spin pure-dephasing rate γ is added in the S_z product basis. Taking $\gamma \to \infty$ at fixed τ measures Zeno freezing rather than a classical limit, because the golden-rule transfer rate scales as $|H_{ij}|^2/\gamma$; we therefore hold rate $\times\, \tau$ fixed by scaling $\tau \propto \gamma$, whereupon the capacity saturates.

TABLE S4. Incoherent limit at matched hopping rate (six seeds).

γ (MHz)	τ (ns)	IPC
0 (coherent)	50	5.421
10^3	10^3	0.999
5×10^3	5×10^3	0.998
2×10^4	2×10^4	0.998

The saturation value 1.00 is the memoryless floor, so the coherence fraction is $(5.421 - 0.999)/5.421 = 0.82$.

S6. CLASSICAL BASELINES

Tanh echo-state networks (spectral radius 0.9, input scale 0.6) are driven by the same input and scored with the same estimator (eight seeds, $L = 1200$).

TABLE S5. The quantum reservoir against matched classical reservoirs.

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.593 ± 0.191
ESN, 5 nodes (5 linear)	4.552 ± 0.325
ESN, 5 nodes (12: 5 linear + 7 quadratic)	9.414 ± 0.860
ESN, 12 nodes (12 linear)	10.366 ± 0.590

Matched by physical unit the quantum system is ahead; matched by readout dimension it is behind. We report both and claim no advantage. Which accounting is fair depends on a cost model for non-commuting measurements against multiplications that we do not have.

S7. TURNOVER CLOCK AND THE REGISTER-REUSE CRITERION

The memory horizon is MC times the cycle time. Because the nuclei leave in the diamagnetic product, where nuclear relaxation is of order 0.1–1 s, the cycle time is the turnover interval T_d and not the radical-pair lifetime. We insert a pause between turnovers and relax the register with probability $q = 1 - \exp(-T_\mathrm{d}/T_{1n})$, taking $T_{1n} = 1$ s.

TABLE S6. The horizon follows the turnover interval.

T_d	q	MC (5 ch)	MC (8 ch)	horizon (8 ch)
1 μs	10^{-6}	1.948	2.818	5.6 μs
1 ms	10^{-3}	1.948	2.818	2.8 ms
10 ms	10^{-2}	1.947	2.817	28 ms
100 ms	9.5×10^{-2}	1.944	2.800	280 ms

The capacity changes by 0.2% across five decades of turnover interval, so the horizon simply tracks it. Table S7 isolates the requirement that actually binds: how much depolarisation of the register between turnovers the memory tolerates.

TABLE S7. The criterion: memory against register depolarisation q.

q	T_d/T_{1n}	MC (5 ch)	MC (8 ch)
0.00	0	1.984	2.890
0.30	0.36	1.958	2.667
0.50	0.69	1.927	2.288
0.70	1.20	1.788	2.030
0.90	2.30	1.218	1.977
1.00	∞	1.002	1.002

Memory is essentially intact while the turnover interval is shorter than the nuclear relaxation time ($q \lesssim 0.5$) and collapses to the memoryless floor MC = 1.00 when every turnover starts from a fresh molecule.

S8. SPIN PARAMETERS

Each radical carries its own nuclei; they are not shared, which is what a spatially separated pair means. The Hamiltonian therefore contains three hyperfine terms, two on the first radical and one on the second. An earlier version of the code accepted a fourth hyperfine argument that was never added to the Hamiltonian and had no effect on any result; it has been removed rather than left in place.

TABLE S8. Hyperfine assignment at the cryptochrome point. Values are representative isotropic magnitudes of the dominant $\mathrm{FAD}^{\bullet-}$ and $\mathrm{TrpH}^{\bullet+}$ couplings; the full anisotropic tensors are given by Lee *et al.*, J. R. Soc. Interface **11**, 20131063 (2014), and the flavin-radical ENDOR couplings by Schleicher *et al.*, Sci. Rep. **11**, 18395 (2021).

coupling	radical / nucleus	(mT)	(MHz)
A_{e_1a}	$\mathrm{FAD}^{\bullet-}$, ^{14}N5	~ 0.50	14
A_{e_1b}	$\mathrm{FAD}^{\bullet-}$, ^{14}N10	~ 0.18	5
A_{e_2a}	$\mathrm{TrpH}^{\bullet+}$, Hβ1	~ 1.43	40

The engineered operating point uses $A_{e_1a} = 80$, $A_{e_1b} = 40$, $A_{e_2a} = 15$ MHz, $J = 2$ MHz, $B = 1$ mT and $\tau = T_2^e = 50$ ns. The conclusions depend on these magnitudes lying in the few–40 MHz range typical of flavin and tryptophan radicals, not on the precise values; the register criterion of Sec. S7 is independent of the hyperfine assignment altogether.